

Data Cleaning & Visualization Project

Customer Spending Dataset — Cleaning, Analysis & Visual Insights

1. Project Overview

This project works with a raw customer dataset containing missing values, outliers, and duplicate records. The goal is to clean and process the data using Python (Pandas), then visualize key insights using Matplotlib and Seaborn to understand customer demographics, spending behavior, and satisfaction levels.

2. Raw Dataset Summary

The raw dataset contained **310 rows** and **7 columns**, including customer ID, name, age, annual income, monthly spend, city, and satisfaction score.

Column	Missing Values (Before)
Age	15
AnnualIncome	16
MonthlySpend	15
SatisfactionScore	17
Duplicate Records	10

3. Data Cleaning Steps

Step 1 - Removing Duplicates: Identified and removed 10 duplicate customer records based on CustomerID, keeping the first occurrence.

Step 2 - Handling Missing Values: Missing numeric values in Age, AnnualIncome, MonthlySpend, and SatisfactionScore were imputed using the median of each column, which is robust to skew and outliers.

Step 3 - Handling Outliers: Outliers in AnnualIncome and MonthlySpend were detected using the IQR (Interquartile Range) method and capped at the lower/upper bounds ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$). This adjusted 6 values in AnnualIncome and 6 values in MonthlySpend.

Result: The cleaned dataset has 300 rows and 7 columns, with zero missing values and zero duplicate records.

4. Visual Insights

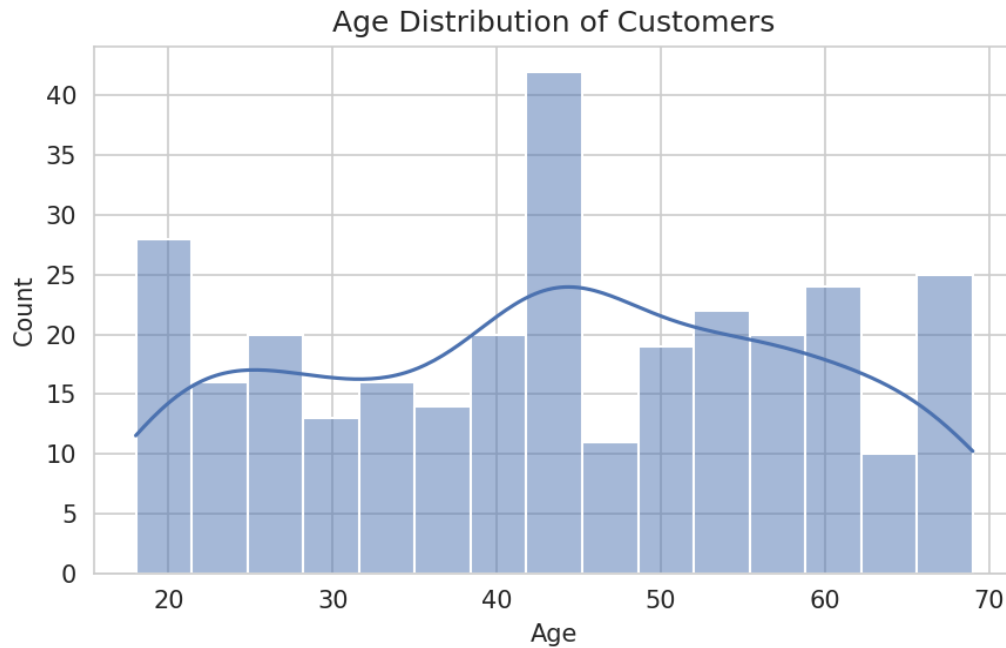


Figure 1: Age distribution shows most customers fall between 25-55 years.

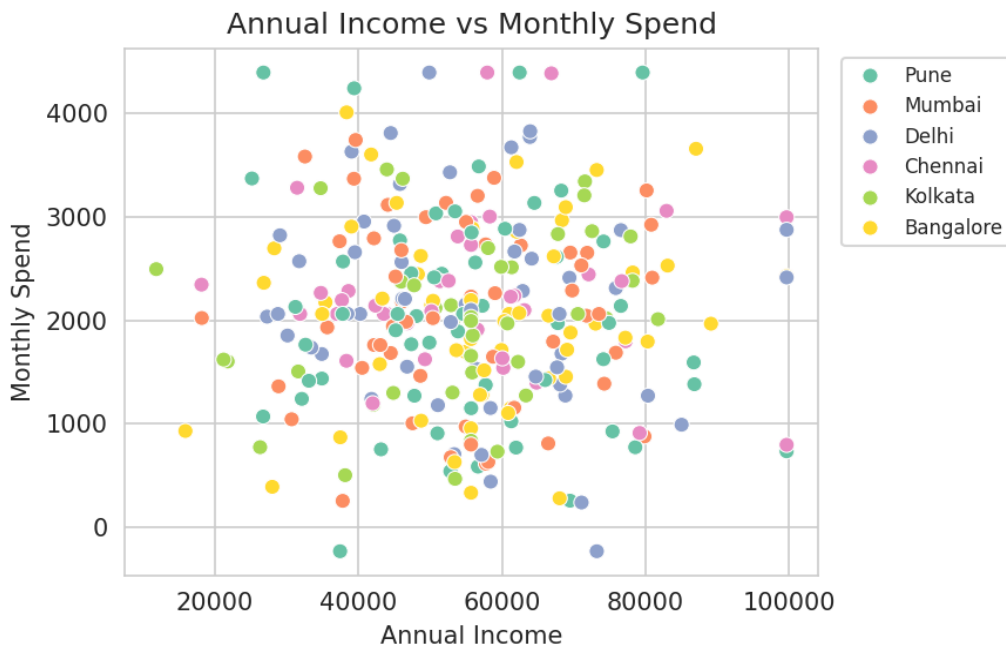


Figure 2: Relationship between annual income and monthly spend across cities.

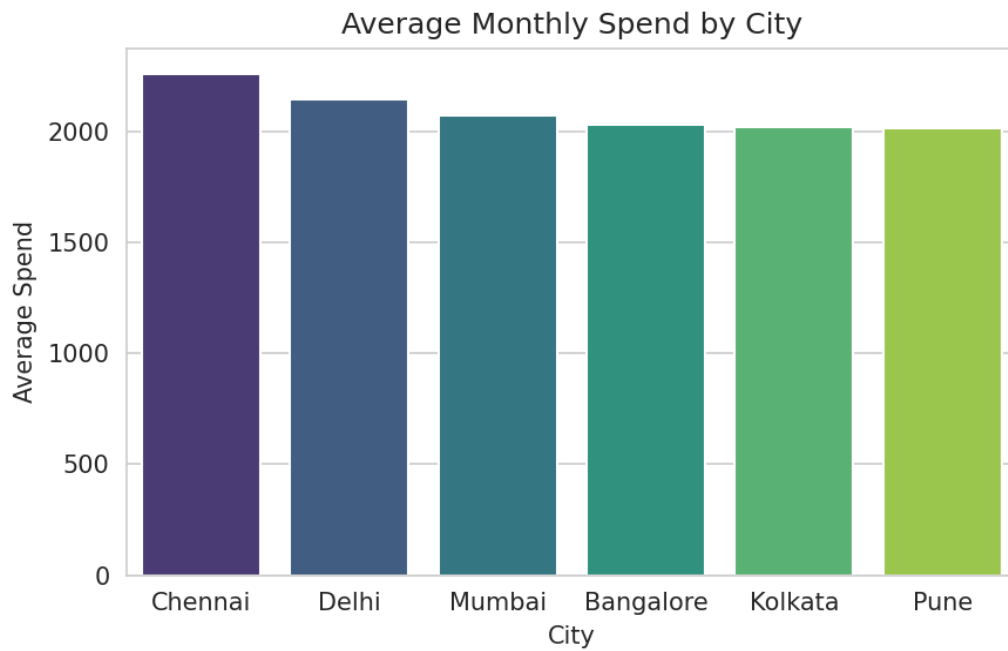


Figure 3: Average monthly spend varies by city.

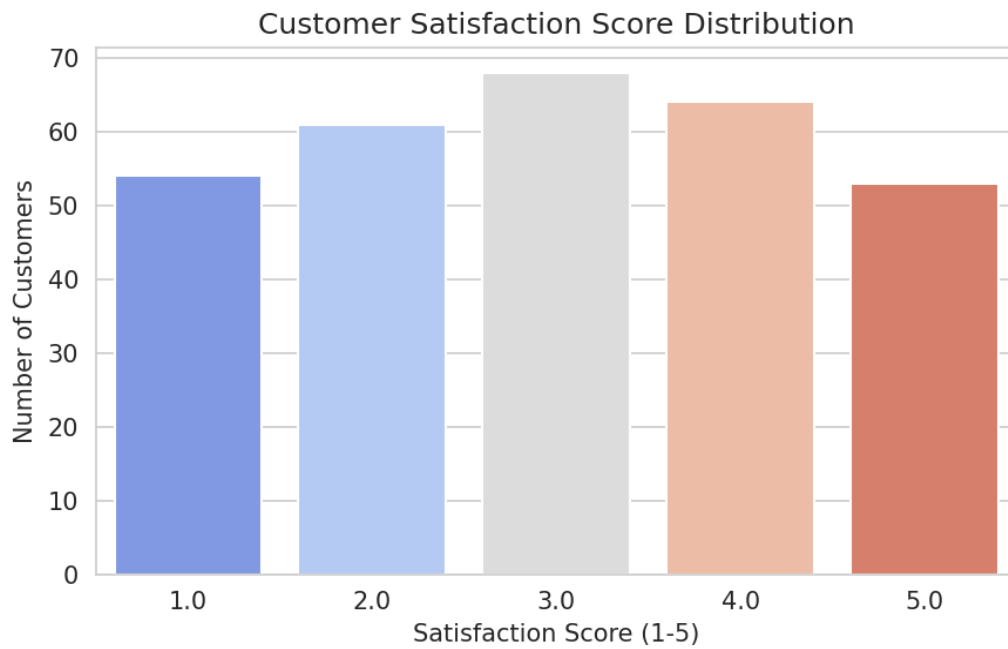


Figure 4: Distribution of customer satisfaction scores (1-5 scale).

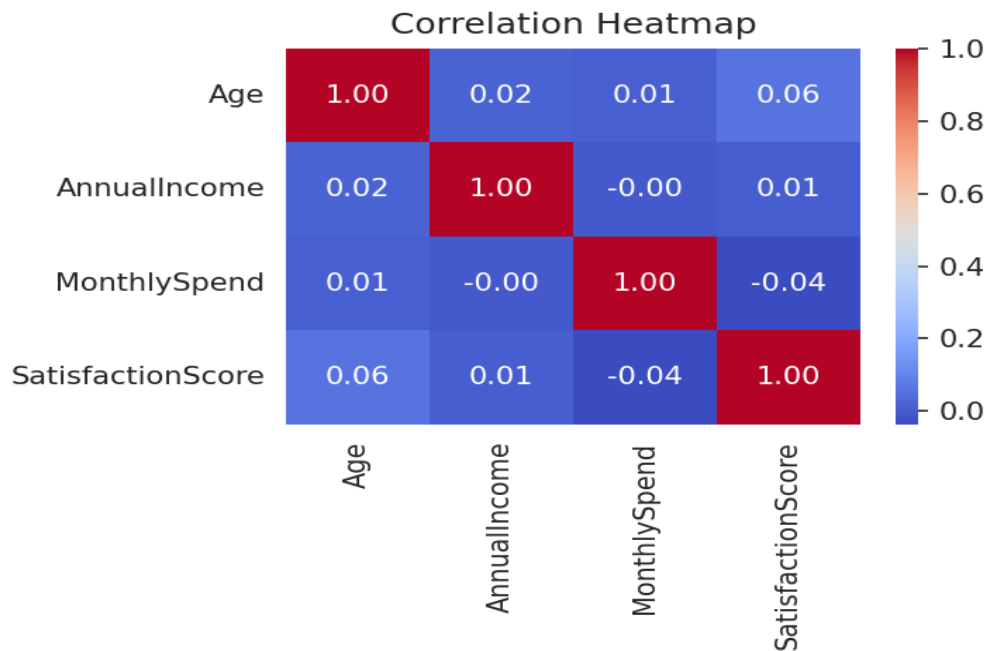


Figure 5: Correlation heatmap between Age, Income, Spend, and Satisfaction.

5. Key Findings

- Customer ages are fairly evenly spread between 25 and 55, with a slight concentration in the 30-45 range.
- No strong linear correlation exists between income and spend, suggesting spending habits are driven by factors beyond income alone (e.g. lifestyle, city).
- Average monthly spend differs noticeably by city, indicating regional spending behavior differences.
- Satisfaction scores are fairly evenly distributed, with no single score dominating - suggesting a mixed customer experience that may warrant further segmentation analysis.
- After cleaning, the dataset is fully consistent (no missing values, no duplicates, outliers capped), making it reliable for further modeling or dashboarding.

6. Expected Outcome / Conclusion

This project demonstrates a complete data preprocessing and visualization workflow: handling missing values and duplicates, treating outliers with the IQR method, and building visual reports using Pandas, Matplotlib, and Seaborn. The resulting clean dataset and visual dashboard provide a solid foundation for deeper analysis, storytelling, and data-driven decision-making.